

DBCLS BioHackathon 2026 report: QPX: Pathway analysis environment for non-model organisms

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[QPX](#)

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Introduction

As part of the DBCLS BioHackathon 2026, we here report the development of pathway analysis environment called Quest for Pathways with eXpression (QPX) designed for non-model organisms.

Results and Discussion

Following BH23 (Pico et al., [2023](#)) and BH25 (Bono et al., [2026](#)), discussions continued at BH26 with the aim of enhancing the usability of QPX. At BH26, together with three new members from Hiroshima University (two undergraduate students and one postdoctoral researcher), we worked on the following points toward the development of pathway data for non-model organisms.

PathLift in use

PathLift is a command line interface (CLI) tool that 'lifts' pathways in Graphical Pathway Markup Language (GPML) from WikiPathways by mapping them to gene IDs in other species, using an ortholog mapping table. It outputs a GPML file in which the GeneProduct nodes from the source species (e.g. human) have been replaced and expanded to candidate gene nodes for the target species. Create new pathways for non-model organisms with public DB ID annotations

Following domestic BioHackathon in Japan (BH26.6), we conducted several projects to generate pathway data for non-model organisms using PathLift.

As an example of cross-species pathway reuse, we used PathLift to map an *Arabidopsis thaliana* mitochondrial complex I pathway to *Oryza sativa* gene candidates (Fig. 1). The resulting GPML retains the pathway layout while displaying the corresponding rice gene IDs.

Figure 1. Arabidopsis-to-rice pathway liftover using PathLift. The N module of mitochondrial complex I is shown before (left) and after (right) mapping to rice gene candidates.

QPX in Practice

We created several annotated pathway diagrams and used them for biological pathway analysis (Table 1).

Table 1: New pathways created in BH26

Pathway	Species	Note
Glycolysis and gluconeogenesis (WP534)	<i>Homo sapiens</i>	Integration of (meta-analysis of) transcriptomes and proteome data
Pentose phosphate, etc	<i>Symplocarpus renifolius</i>	Integration of metabolome and transcriptome data
Food Allergy-Associated Mast Cell Activation Pathway	<i>Homo sapiens</i>	
Mitochondrial complex I	<i>Arabidopsis thaliana</i> <i>Oryza sativa</i>	
Photosynthetic carbon reduction (WP1461)	<i>Arabidopsis thaliana</i>	

Integration of transcriptomes and proteome data

Use case for the integration of multiple omics data was investigated. Differentially abundant proteins (DAPs) in hypoxic stress from proteome and Meta-analysis of public transcriptomes and hypoxic stress were extracted from manuscripts below.

1. Transcriptome: hypoxic stress RNA-seq data (HN-score) collected in “Multi-Omic Meta-Analysis of Transcriptomes and the Bibliome Uncovers Novel Hypoxia-Inducible Genes. DOI: 10.3390/biomedicines9050582” (Ono & Bono, 2021).
2. Proteome: differentially abundant proteins (DAPs) flags in “Proteomic-Based Analysis of Hypoxia- and Physioxia-Responsive Proteins and Pathways in Diffuse Large B-Cell Lymphoma. DOI: 10.3390/cells10082025”

Integration of transcriptome and metabolome data

A minimal set for reproducing the display of gene expression and metabolome data from the thermogenic tissues Hot_F / Hot_P of the Asian skunk cabbage (*Symplocarpus renifolius*) on pathway maps with QPX.

Future work

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Author contributions

Hidemasa Bono: Conceptualization, Writing - original draft; Naoya Oec: System development, Validation; Haruka Tanimoto: Data creation; Tomohiro Taguchi: Data creation; Amane Kurata: Data creation; Ryo Nozu: Validation, Data creation.

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